

EEE-2421: Electrical Drives and Instrumentation

Credit Hours - 2

Segment - 3

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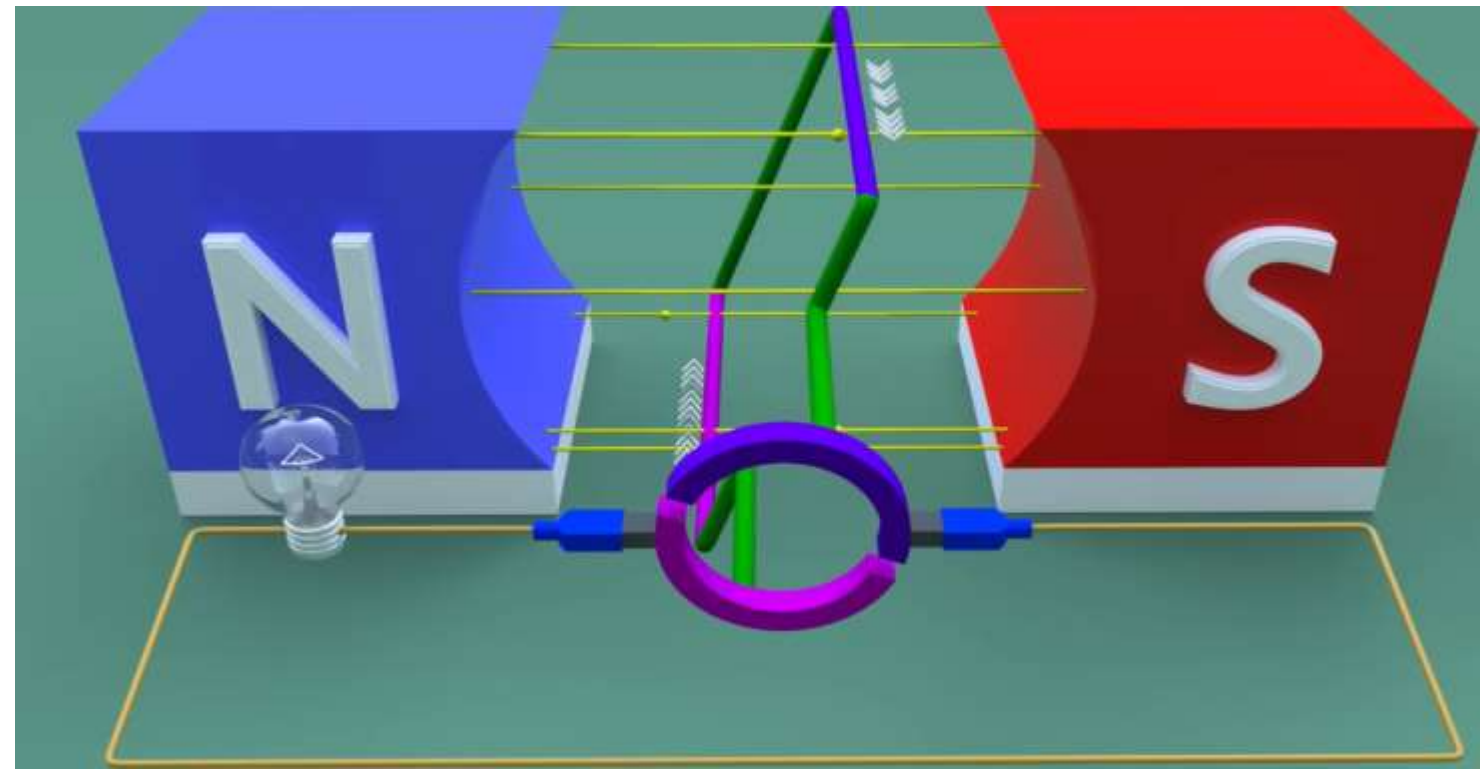
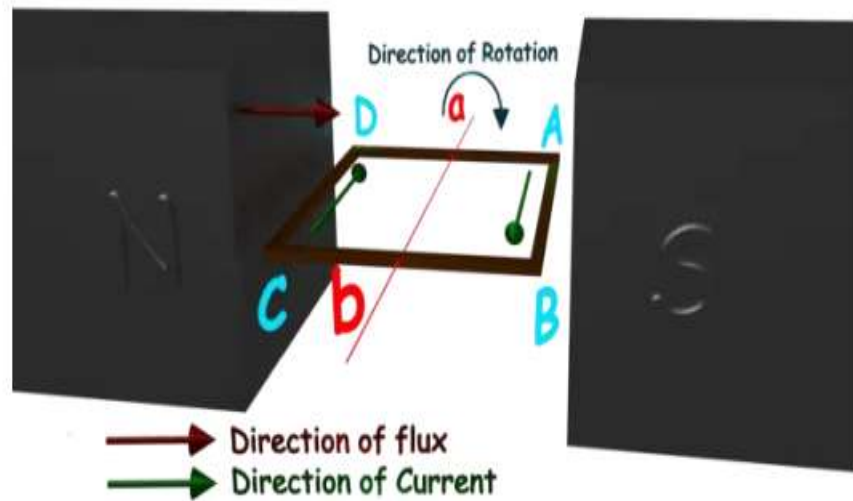
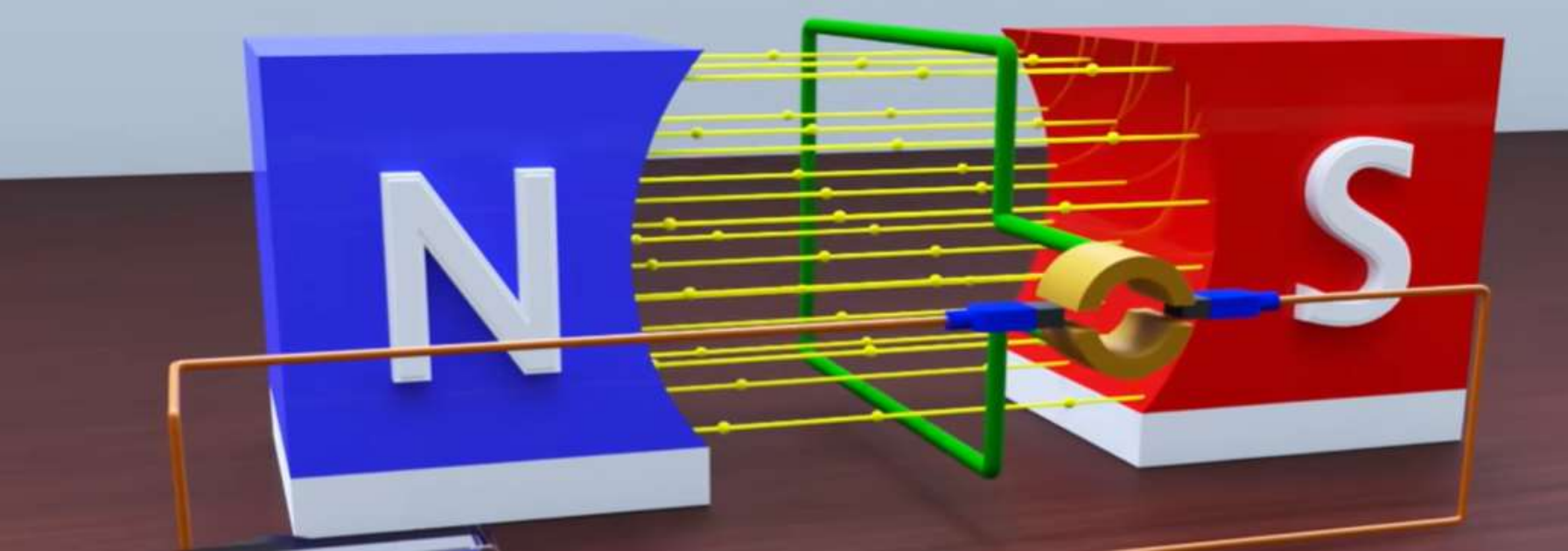
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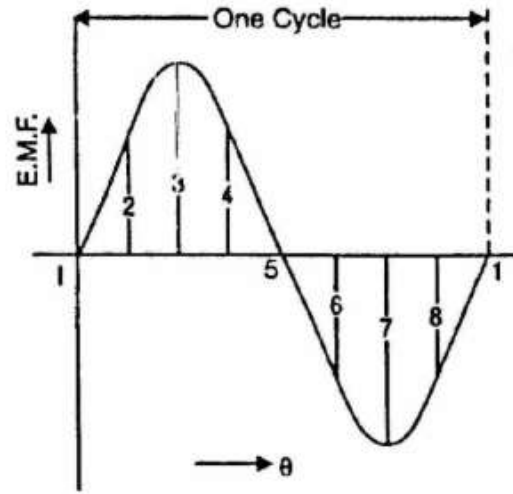
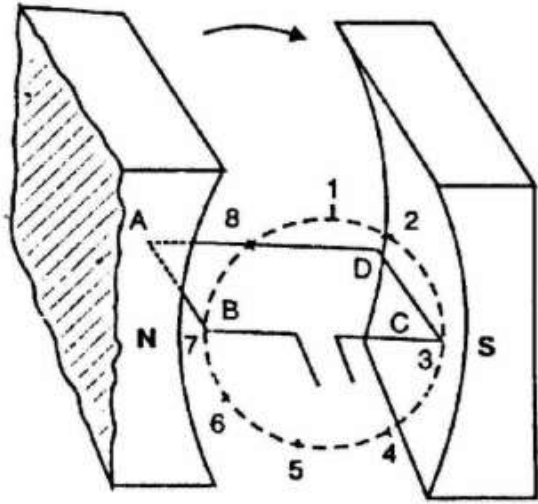
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DC Generator

- There are two types of generators—DC generators and AC generators.
- Both DC and AC generators convert mechanical power to electrical power.
- A DC generator produces direct power, while an AC generator produces alternating power.
- Both of these generators produce electrical power based on the principle of Faraday's law of electromagnetic induction.



DC Generator Working Principle



Consider a single turn loop ABCD rotating clockwise in a uniform magnetic field with a constant speed as shown in Fig. As the loop rotates, the flux linking the coil sides AB and CD changes continuously. Hence the e.m.f. induced in these coil sides also changes but the e.m.f. induced in one coil side adds to that induced in the other.

- (i) When the loop is in position no. 1 [See Fig. 1.1], the generated e.m.f. is zero because the coil sides (AB and CD) are cutting no flux but are moving parallel to it.
- (ii) When the loop is in position no. 2, the coil sides are moving at an angle to the flux and, therefore, a low e.m.f. is generated as indicated by point 2 in Fig 1.2.
- (iii) When the loop is in position no. 3, the coil sides (AB and CD) are at right angle to the flux and are, therefore, cutting the flux at a maximum rate. Hence at this instant, the generated e.m.f. is maximum as indicated by point 3 in Fig. (1.2).
- (iv) At position 4, the generated e.m.f. is less because the coil sides are cutting the flux at an angle.
- (v) At position 5, no magnetic lines are cut and hence induced e.m.f. is zero as indicated by point 5 in Fig. (1.2).
- (vi) At position 6, the coil sides move under a pole of opposite polarity and hence the direction of generated e.m.f. is reversed. The maximum e.m.f. in this direction (i.e., reverse direction, See Fig. 1.2) will be when the loop is at position 7 and zero when at position 1. This cycle repeats with each revolution of the coil.

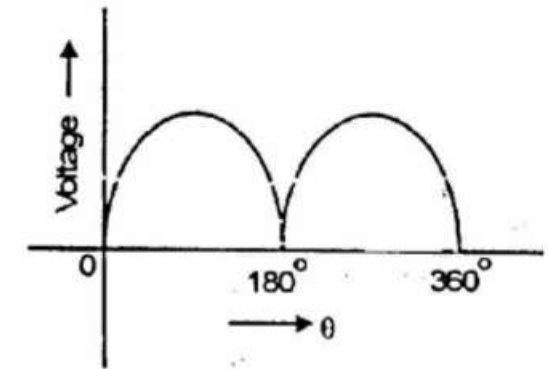
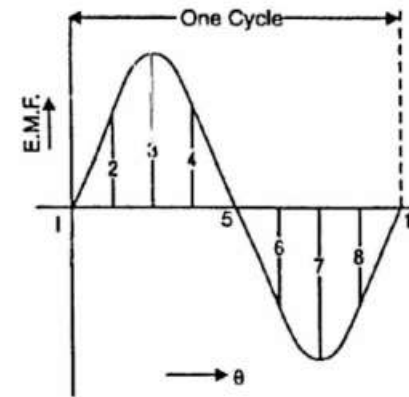
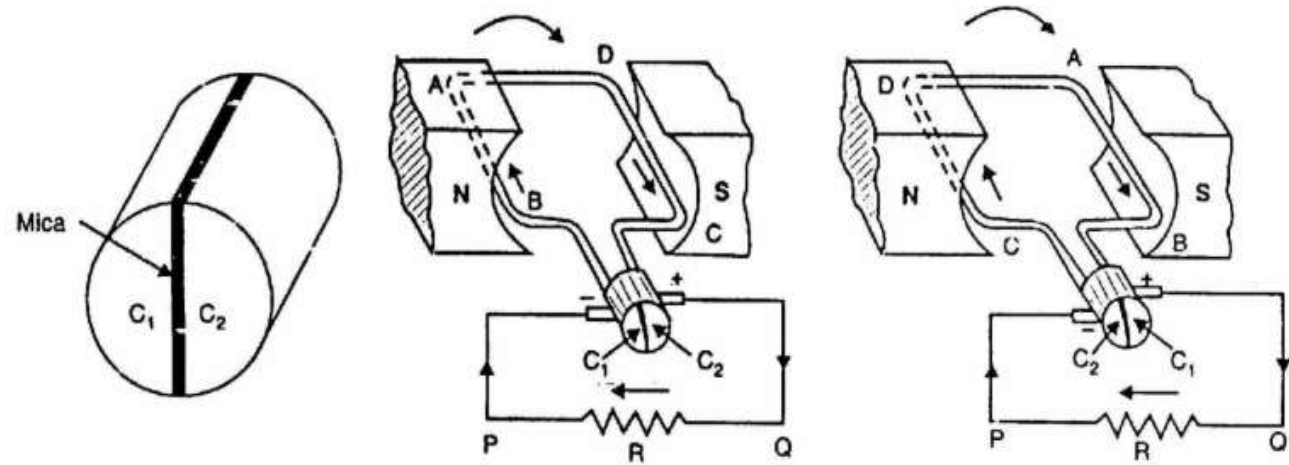
Action of Commutator

Commutation is the process of converting the AC voltages and currents in the rotor of a DC machine to DC voltages and currents at its terminals.

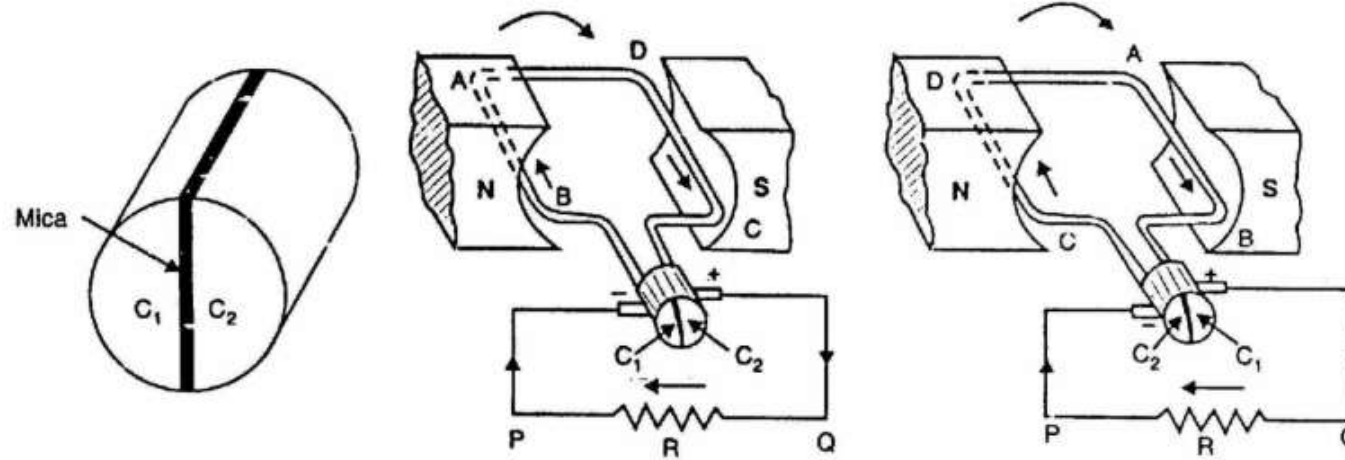
When the voltage reverses in a loop, the connections of the loop are also switched to keep the polarity of the terminal voltage the same.

Two stationary carbon brushes rest on the commutator and lead current to the external load. With this arrangement, the commutator at all times connects the coil side under S-pole to the +ve brush and that under N-pole to the -ve brush.

Thus, AC is converted into pulsating DC.



Faraday's Law

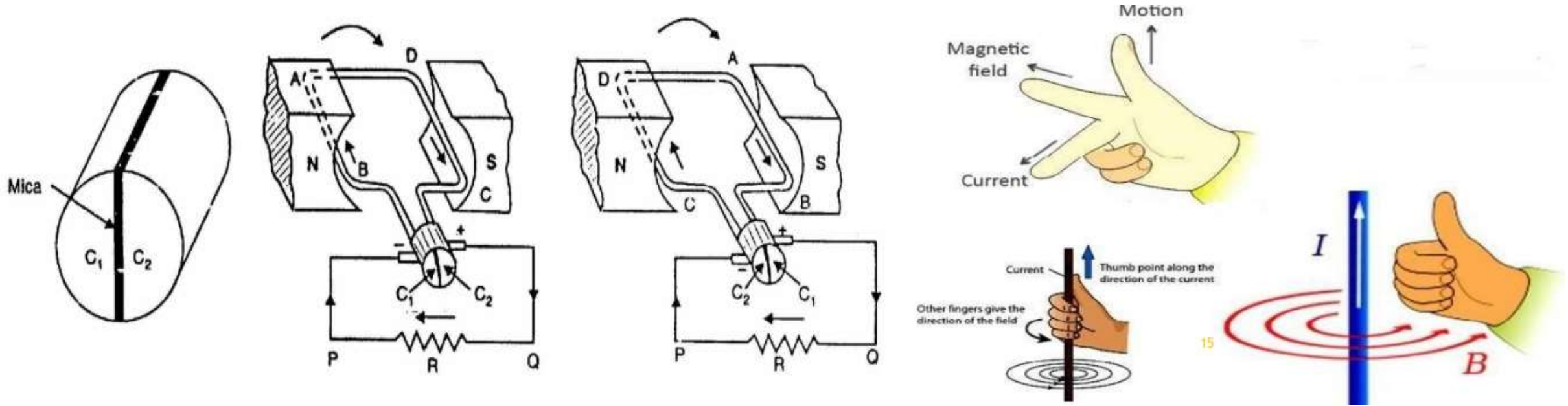


First Law: Whenever a conductor is rotated in magnetic field emf is induced in it. If the conductor is closed, current will flow through it which is known as induced current.

Second Law: Second law states that induced emf is equal to the rate of change of linkage or flux cut. This EMF will cause a current to flow if the conductor circuit is closed.

$$EMF = -N \frac{d\Phi}{dt}$$

Flemming's Right Hand Rule

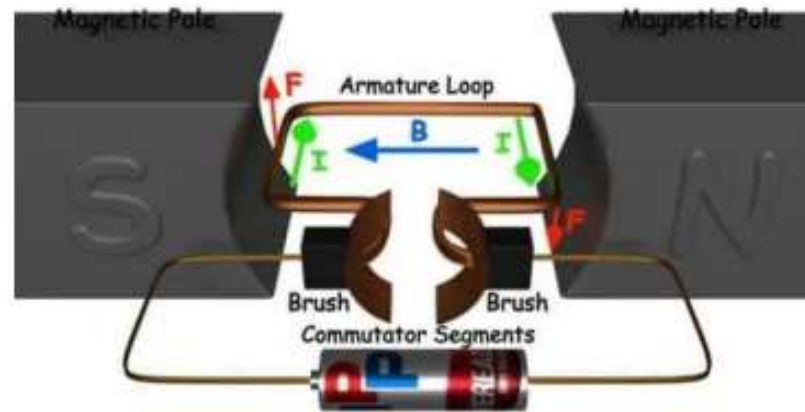


If you stretch the thumb, index finger, and middle finger of your right-hand perpendicular to each other:

- Thumb indicates the direction of motion of the conductor,
- Index finger indicates the direction of the magnetic field, i.e., N – pole to S – pole, and
- Middle finger indicates the direction of flow of current through the conductor.

DC Motor Working Principle

To understand the **operating principle of DC motor** we need to first look into its single loop constructional feature.



A machine that converts d.c. power into mechanical power is known as a d.c. motor. Its operation is based on the principle that when a current carrying conductor is placed in a magnetic field, the conductor experiences a mechanical force. The direction of this force is given by Fleming's left hand rule and magnitude is given by;

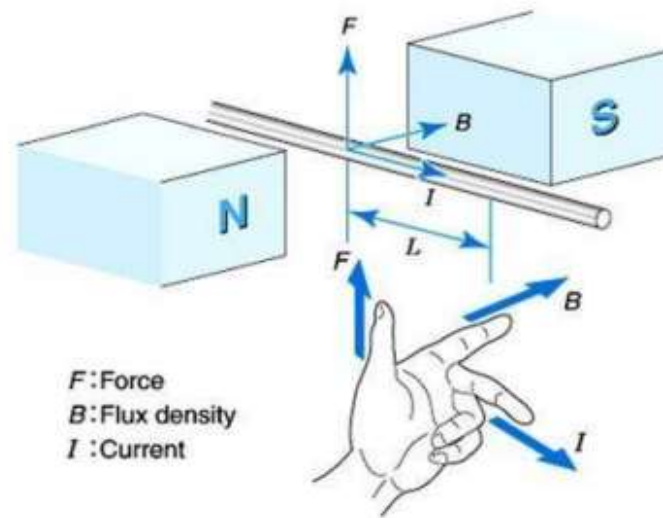
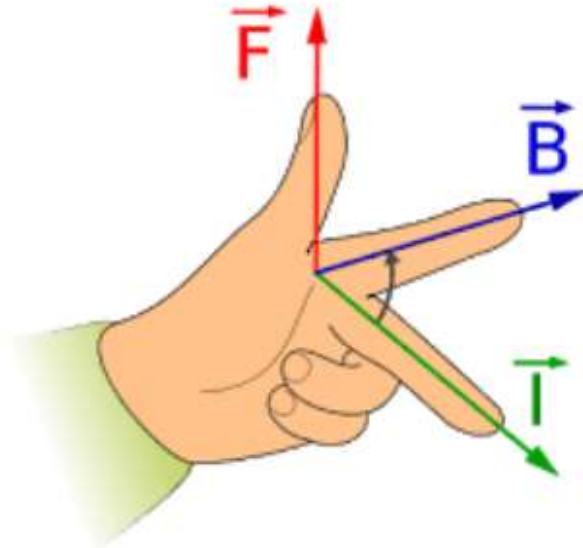
$$F = BIl$$

By applying Fleming's left hand rule, it is clear that force on each conductor is tending to rotate the armature in anticlockwise direction. All these forces add together to produce a driving torque which sets the armature rotating. When the conductor moves from one side of a brush to the other, the current in that conductor is reversed and at the same time it comes under the influence of next pole which is of opposite polarity.

Consequently, the direction of force on the conductor remains the same. Basically, there is no constructional difference between a d.c. motor and a d.c. generator. The same d.c. machine can be run as a generator or motor.

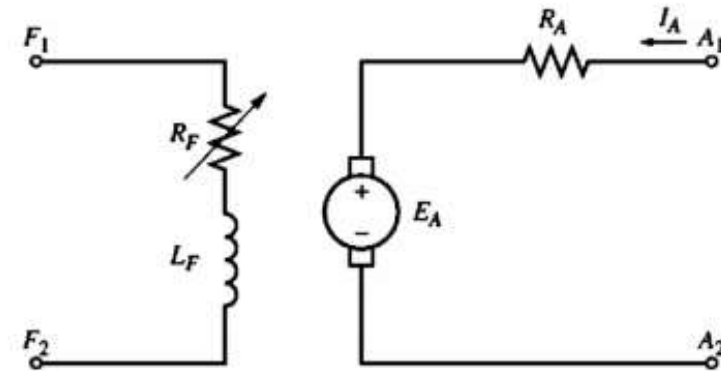
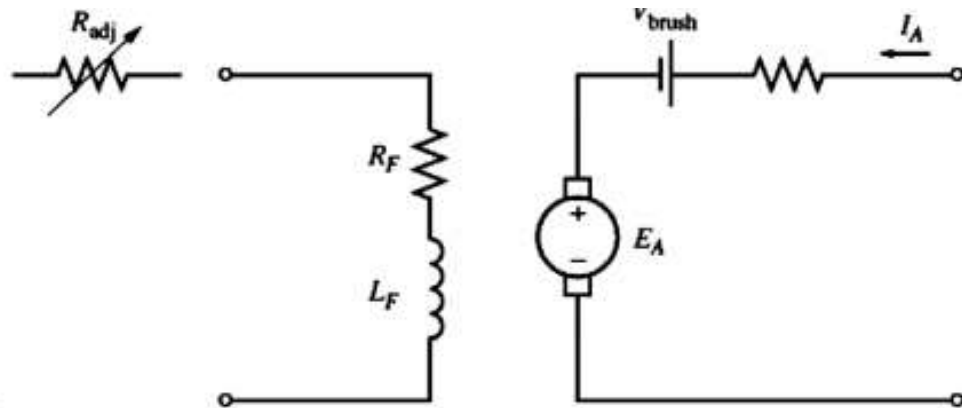
Flemming Left Hand Rule

- Hold out your left hand with the index finger, middle finger and thumb at the right angle to one another.
- If the index finger represents the direction of the field and the middle finger represents that of the current, then thumb gives the direction of the force.



Equivalent Circuit of DC Motor

The equivalent circuit of a dc motor is shown in Figure . In this figure, the armature circuit is represented by an ideal voltage source E_A and a resistor R_A . This representation is really the Thevenin equivalent of the entire rotor structure, in-circuit. The brush drop voltage is often only a very tiny fraction of the generated voltage in a machine. Therefore, in cases where it is not too critical, the brush drop voltage may be left out or approximately included in the value of R_A . Also, the internal resistance of the field coils is sometimes lumped together with the variable resistor, and the total is called R_F (see Figure



Numerical Problems

Example 26.3. A shunt generator delivers 450 A at 230 V and the resistance of the shunt field and armature are $50\ \Omega$ and $0.03\ \Omega$ respectively. Calculate the generated e.m.f.

Solution. Generator circuit is shown in Fig. 26.46.

Current through shunt field winding is

$$I_{sh} = 230/50 = 4.6\text{ A}$$

Load current $I = 450\text{ A}$

$$\begin{aligned}\therefore \text{Armature current } I_a &= I + I_{sh} \\ &= 450 + 4.6 = 454.6\text{ A}\end{aligned}$$

Armature voltage drop

$$I_a R_a = 454.6 \times 0.03 = \mathbf{13.6\text{ V}}$$

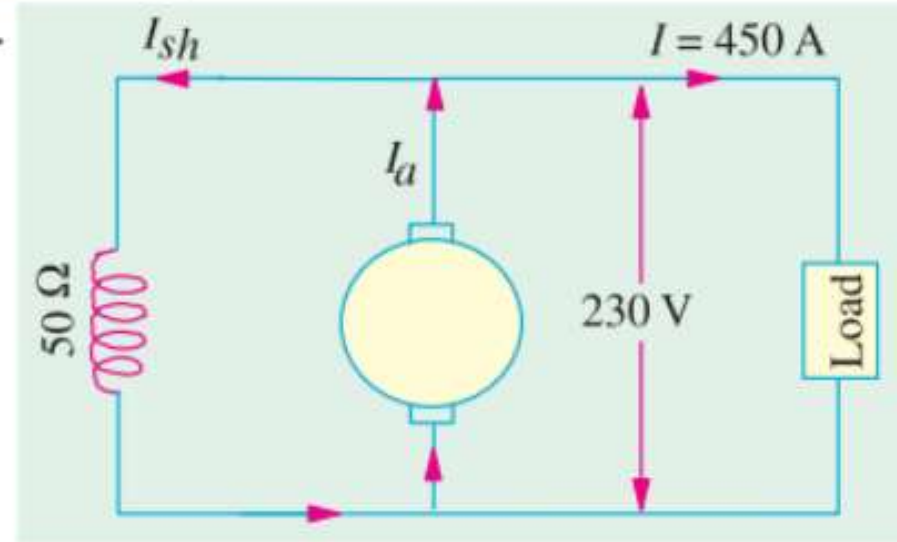


Fig. 26.46

$$\begin{aligned}\text{Now } E_g &= \text{terminal voltage} + \text{armature drop} \\ &= V + I_a R_a\end{aligned}$$

$\therefore$ e.m.f. generated in the armature

$$E_g = 230 + 13.6 = \mathbf{243.6\text{ V}}$$

Numerical Problems

Example 29.21. A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8A. The total armature and shunt field resistances are respectively 0.2 Ω and 250 Ω . Calculate the speed when loaded and taking 50 A. Assume the flux to be constant. (Elect. Engg. A.M.Ac. S.I. June 1991)

Solution. Formula used : $\frac{N}{N_0} = \frac{E_b}{E_{b0}} \times \frac{\Phi_0}{\Phi}$; Since $\Phi_0 = \Phi$ (given); $\frac{N}{N_0} = \frac{E_b}{E_{b0}}$

$$I_{sh} = 250/250 = 1 \text{ A}$$

$$E_{b0} = V - I_{a0} R_a = 250 - (7 \times 0.2) = 248.6 \text{ V}; E_b = V - I_a R_a = 250 - (49 \times 0.2) = 240.2 \text{ V}$$

$$\therefore \frac{N}{1000} = \frac{240.2}{248.6}; N = 9666.1 \text{ r.p.m.}$$